CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

V Semester of BSc Biological Sciences Examination November 2019

Course code & Name: BT506 Animal Biotechnology

07.11.2019 Thursday 01:30 PM TO 02:00 PM Maximum Marks: 20

MCQ

Important Instructions:

- Tick the correct answer and it should be written in question paper itself.
- Use of non-programmable calculator is allowed.

Q - I Choose the correct answer for the following questions. **20**

1. Select the wrong combination

(i) Aqua culture/ Blue Biotechnology
(ii) Soil remediation/ White Biotechnology
(iii) Fuel Production/ Green Biotechnology
(iv) Gene Therapy/ Red Biotechnology

2. A gene of interest introduced into an organism by artificial means is called

(i) Marker gene
(ii) Transgene
(iii) Site-specific gene
(iv) Recombinant gene

3. Inorganic salts in a culture media serves as

(i) Osmotic pressure regulator
(ii) pH indicator
(iii) Buffer
(iv) All of the above

4. Cryopreservation is used to preserve

(i) Semen
(ii) Body tissues
(iii) Early embryos
(iv) All of the above

5. Best method to produce a large number of offspring from livestock with desirable traits is

(i) IVF
(ii) AI
(iii) Natural breeding
(iv) All of the above

6. Which of the following is not a product of marine biotechnology

(i) PUFA
(ii) Agar
(iii) Algins
(iv) PEG

7. Subhash Mukhopadhyay and Robert Edwards pioneered in the field of

(i) IVF
(ii) AI
(iii) Menarche
(iv) Menopause

8. HeLa is a

(i) Breast cancer cell line
(ii) Cervical cancer cell line
(iii) Type of brain stem cell
(iv) Type of lung stem cell

9. mAbs are

(i) heterogenous antibodies produced from single clone of plasma cells
(ii) heterogenous antibodies produced from polyclonal plasma cells
(iii) homogenous antibodies produced from single clone of plasma cells
(iv) homogenous antibodies produced from polyclonal plasma cells

10. Identify the first genetically modified bacterium that could degrade the hydrocarbon present in oil spills

(i) *Pseudomonas putida*
(ii) *Pseudomonas pneumonia*
(iii) *Pseudomonas aeruginosa*
(iv) *Bacillus subtilis*

11. The cryoprotectant used to preserve cell lines is

(i) DMSO (ii) Sucrose (iii) HEPES (iv) PEG

12. The Sheep "Dolly" was developed by

(i) transferring the nucleus of the udder cell to nucleated oocyte
(ii) transferring the nucleus of mammary gland stem cell in enucleated oocyte
(iii) transferring the nucleus of the mammary gland cell to enucleated oocyte
(iv) transferring the nucleus of oocyte in the udder cell

13. The spatial expression pattern of cre transgene in Cre mice depends on

(i) *loxp* orientation (ii) *cre* gene (iii) *cre* promoter (iv) All of the above

14. For routine cell culture, the most common laboratory safety level is

(i) I (ii) II (iii) III (iv) IV

15. Vitrification is associated with

(i) Thawing of cells
(ii) Freezing of cells
(iii) Maintaining physiologic state of the cells in culture
(iv) Apoptosis and autophagy

16. Antibody producing hybridoma is a fusion of

(i) Myeloma cell and T cell
(ii) Myeloma cell and Macrophage cell
(iii) Myeloma cell and NK cell
(iv) None of the above

17. Location of loxP sites in a Cre/Lox model is

(i) Adjacent to the Cre locus
(ii) At the random sites of the nuclear genome
(iii) On either side of the gene to be knocked out
(iv) At the random sites of the mitochondrial genome

18. Xenotransplantation is grafting of tissue/ organ between

(i) Two different species
(ii) Two individuals of the same species
(iii) Different body parts of an individual
(iv) None of the above

19. Low amount of oxygen to the cells result in accumulation of _____ in culture media

(i) Lactic acid
(ii) Ethanol
(iii) ATPs
(iv) All of the above

20. ICSI is

(i) Intracytoplasmic sperm injection
(ii) Intracytoplasmic sperm introduction
(iii) Intercytoplasmic sperm injection
(iv) Intercytoplasmic sperm introduction

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V Semester of BSc Biological Sciences Examination November 2019

Course code & Name: BT506 Animal Biotechnology

07.11.2019 Thursday 02:01 PM TO 04:30 PM Maximum Marks: 50

Instructions:

1. **Section I and II must be attempted in TWO ANSWER SHEETS**
2. Make suitable assumptions and draw neat figures wherever required.
3. Use of non-programmable calculator is allowed.
4. Show necessary calculations.

SECTION -I

Q - II Attempt ANY FOUR **20**

1. Define animal biotechnology and its scopes. Write down six major applications of animal biotechnology?
2. Discuss the applications of animal cell culture and the importance of various equipment and materials used in cell line culture.
3. What is a cell line? Discuss the differences between tumor cell, primary cell and immortalized cell?
4. Describe various strategies for the generation of transgenic animals with a special focus on microinjection.
5. Describe the following:
   (i) Monoclonal and polyclonal antibodies
   (ii) Embryo twinning and embryo sexing

SECTION – II

Q - III Attempt ANY SIX **30**

1. Discuss the utility of compounds obtained from marine sources and the importance of marine animals in animal biotechnology.
2. What is cryopreservation and its importance? Describe the methods of oocyte and/ or embryo cryopreservation.
3. Describe SCNT for animal cloning.
4. How AI and IVF differ from each other? Discuss their applications.
5. What is hybridoma technology? Discuss its importance.
6. What is a Cre-Lox mouse model? Discuss its importance.
7. Write down a strategy (schematic flow chart) for temporal and spatial knocking out of a gene “X” in brain cells of a young mice using cre-loxP system.